INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

SWAGING MACHINE, ROTARY, FLOOR- MOUNTED

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers floor-mounted rotary swaging machines and accessories intended for use in pointing, reducing diameters, and forming shapes on tubing and solid stock and swaging fittings to cables and hoses.

2. CLASSIFICATION. The rotary swaging machines shall be of the following types and sizes. Swaging machine sizes are shown in tables I, II, and III. The type and size to be furnished shall be as specified (see 7.2(b)).

Type I - Rotary die swaging machine (See Table I)

Type II - Stationary die swaging machine (See Table II)

Type III - Rotary die-closing swaging machine (See Table III)

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data that may improve this document should be sent to: STDZNMGT@dla.mil Defense Logistics Agency-Aviation, ATTN: VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616. Since contact information can change, you may want to verify the currency of this address information using assist database at https://assist.dla.mil.

FSC 3446

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TABLE I. Sizes of type I rotary die swaging machines.

Size	Diametral capacity, inches	
	Solids	Tubing
1	1/16	¼
2	1/8	5/16
3	3/8	¾
4	1/2	1
5	5/8	1-3/4
6	1	2
7	1-3/8	3
8	2	4
9	3	6

TABLE II. Sizes of type II stationary die swaging machines.

Size	Diametral capacity, inches	
	Solids	Tubing
1	13/32	1
2	3/4	2-1/4
3	1-1/2	3-3/8
4	2	4-1/2
5	2-1/2	6

TABLE III. Sizes of type III die-closing swaging machines.

Size	Diametral capacity, inches		
	Solids	Tubing	Cable
1	3/16	1/2	1/16
2	3/8	3/4	¼
3	1/2	1	5/16
4	1	2	¾
5	1-1/2	3	1

NOTE: Machine size and capacity for each type shall not be less than the stated requirements of the size ordered nor greater than the next larger size as shown in table I, II, or III.

3. SALIENT CHARACTERISTICS

3.1 General requirements. The machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The swaging machine shall include, as a minimum, the following components:

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3.2.1 Base. The base shall support the other elements of the machine and shall incorporate materials, shape, and mass to minimize adverse vibration and deflection. The base shall provide a means for leveling and anchoring the machine.

3.2.2 Electric-motor drive. The rotary swaging machine flywheel shall be driven by an electric motor through a belt drive system. Unless otherwise specified (see 7.2(c)), motor horsepower, speed, and drive system shall be that considered by the manufacturer as standard.

3.2.3 Flywheel. The flywheel shall be a casting of iron or steel and dynamically balanced. The design and mass of the flywheel shall store and deliver energy required to perform all operations within machine capacity without imposing overload on the electric motor. On a type I or III machine, the flywheel shall be mounted directly on the spindle. On a type II machine, the roll cage shall be mounted within the flywheel.

3.2.4 Spindle. A hollow steel spindle shall be mounted in the head on antifriction bearings of a size, type, and precision suitable for the application and designed to withstand repeated axial loads. Unless otherwise specified (see 7.2(d)), the spindle hole diameter shall be that considered by the manufacturer as standard and meet or exceed the machine tubing diametral capacity.

3.2.5 Dies. Unless otherwise specified (see 7.2(e)), the swaging machine shall be the two-die type. If required, dies shall be provided as specified (see 7.2(f)).

3.2.5.1 Cable terminal dies. If required, the type III swaging machine shall include dies to swage the following standard cable terminals as specified (see 7.2(g)):

a. Fork end.
b. Eye end.
c. Single shank ball.
d. Double shank ball.

3.2.6 Roll cage. A roll cage or rack assembly containing steel rollers in recesses shall be mounted in an inside or head ring. On a type I and III machine, the roll cage or rack shall remain free to impart striking force to the rotating hammers. On a type II machine, the roll cage or rack shall revolve with the rotating head to impart striking force to the stationary hammers.

3.2.7 Die-closing device. On a type III machine only, a die-closing device shall be provided. The die-closing device shall consist of hydraulically operated wedges interposed between the hammers and dies, allowing the dies to remain open while the machine is running.

3.2.8 Manual controls. Unless otherwise specified (see 7.2(h)), the following manual switches, as a minimum, shall be provided:

a. Power on/off switches.
b. Hydraulic system power on/off switches (type III machine only).

3.2.9 Coolant system. Unless otherwise specified (see 7.2(i)), the rotary swaging machine shall be flood cooled. As a minimum, the coolant system shall consist of a reservoir, pump, strainer, and all necessary piping. The system shall be equipped with a method to control the rate of coolant flow from the reservoir and to direct coolant flow to the grinding area. When specified (see 7.2(j)), an

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optional coolant filtration system shall be provided.

3.2.10 Hydraulic system. On a type III machine, the hydraulic system shall conform to the requirements of International Organization for Standardization (ISO) ISO-4413. The hydraulic system shall be complete and include all pumps, valves, piping, cylinders, and controls. Means shall be provided to prevent damage to the components from overpressure. A filter system shall be provided to ensure delivery of clean fluid.

3.2.11 Electrical system. Unless otherwise specified (see 7.2(k)), the electrical system shall conform to the American National Standards Institute/National Fire Prevention Association (ANSI/NFPA) 79. The machine shall have dual 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.2(l)).

3.2.12 Electric motors. The rotary swaging machine drive motor shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG-1. The drive motor(s) shall be totally enclosed and fan cooled (TEFC) or open dripproof (ODP) as specified (see 7.2(m)).

3.2.13 Optional accessories. Optional accessories shall be provided as specified (see 7.2(n)).

3.3 Performance. Unless otherwise (see 7.2(o)), the rotary swaging machine shall meet the performance requirements shown in table IV when tested in accordance with section 5.3 through 5.3.5.

TABLE IV. Performance requirements.

Type	Swaging requirements
I	Pointing, forming, tapering, and reducing the diameter on tubing, wire, and malleable solids.
II	Pointing and tapering square and rectangular entry cross sections. Forming fluted shapes, splines, blade ends, and other cross sections from round, square, or rectangular stock.
III	Assembling cable terminals and other fittings to cables. Locking bands to pins or shafts. Forming center reductions between shoulders. Pointing and reducing round stock.

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(p)).

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.7. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910. If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(q)).

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Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(r)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(s)), the nameplate shall contain the following information:

a. Nomenclature.
b. Manufacturer's name.
c. Serial number.
d. Machine model designation.
e. Power input (volts, total amperes, phase, frequency).
f. Short-circuit/over-current rating.
g. Contract number or order number.
h. National stock number.
i. Date of manufacture.

3.8 Lubrication plate or chart. When specified (see 7.2(t)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application.
b. Servicing interval.
c. Type of lubricant(s) with SAE number or lubricant identifier.

3.9 Hydraulic fluid plate or chart. When specified (see 7.2(u)), a hydraulic fluid plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Servicing interval.
b. Type of fluid with SAE number or fluid identifier.
c. Type of filter.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products shall meet the salient characteristics of this commercial item description; conform to the manufacturer's own drawings, specifications, standards, and quality assurance practices; and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

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5.2 Inspection. The rotary swaging machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(v)), the rotary swaging machine shall be tested in accordance with sections 5.3.1 through 5.3.5.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(w)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(x)).

5.3.2 Operational test. The rotary swaging machine shall be operated in accordance with the manufacturer's standard operational test procedures. The rotary swaging machine shall be operated at no load for not less than one hour. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Performance tests. No less than five tubes or solids for the type and size of machine specified, equal to the maximum capacity of the machine, shall be swaged to the shape of the dies requested by the government. The type III machine shall be operated to assemble no less than five fittings to cables. Complete swaging of the assemblies shall be accomplished without distortion of the fittings. All performance tests shall be completed without making any adjustments to or performing any maintenance on the machine.

5.3.4 Noise level tests. Noise level measurements shall be made in accordance with section 8.4.2 of ANSI S1.13 to ensure conformance with noise limits set by OSHA 29 CFR PART 1910.95, Occupational Noise Exposure.

5.3.5 Optional tests. Optional tests shall be conducted as specified (see 7.2(y)).

5.4 Acceptance. Unless otherwise specified (see 7.2(z)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the performance tests shall be compared with the requirements shown in table IV. Failure of the machine to meet the requirements shown in table IV shall be cause for rejection.

6. PACKAGING. Unless otherwise specified (see 7.2(aa)), preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Sources of referenced documents.

7.1.1 Government documents. Copies of Federal documents may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Electronic copies of may be obtained from https://assist.dla.mil/.

Occupational Safety and Health Administration (OSHA)

OSHA 29 CFR Part 1910 Occupational Safety and Health Standards for General Industry

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Applications for copies should be sent to the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210. Electronic copies of OSHA standards may be obtained from https://www.osha.gov/.

Federal Acquisition Regulation (FAR)

FAR Paragraph 23.403 Use of Recovered Materials and Biobased Products

The FAR may be obtained from the US Government Printing Office 732 North Capitol Street N.W., Washington, DC 20401. Electronic copies of OSHA standards may be obtained from http://www.acquisition.gov/far/.

7.1.2 Industry standards. Copies of industry standards referenced in this CID may be obtained from the following addresses:

American National Standards Institute (ANSI)

ANSI B11.7	Machine Tools - Cold Headers and Cold Formers, Safety Requirements for Construction, Care, and Use
ANSI S1.13	Measurement of Sound Pressure Levels in Air

Applications for copies should be sent to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036. Electronic copies of ANSI standards may be obtained from http://www.ansi.org/.

National Electrical Manufacturers Association (NEMA)

NEMA MG-1 Motors and Generators

Applications for copies should be sent to the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209. Electronic copies of NEMA standards may be obtained from http://www.nema.org/.

National Fire Protection Association (NFPA)

ANSI/NFPA 79 Electrical Standard for Industrial Machinery

Applications for copies should be sent to the National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101. Electronic copies of NFPA standards may be obtained from http://www.nfpa.org/.

International Organization for Standardization (ISO)

ISO-4413	Hydraulic Fluid Power – General rules and safety requirements for systems and their components

Applications for copies should be sent to the, Customer Service Division, 545 Washington Boulevard, Jersey City, NJ 07310-1686. Electronic copies of NEPA standards may be obtained from http://www.iso.org/.

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7.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type and size of rotary swaging machine required (see 2).
c. Motor horsepower, speed, and drive system, if different (see 3.2.2).
d. Spindle hole diameter, if different (see 3.2.4)
e. Number of dies, if different (see 3.2.5).
f. Dies, as specified (see 3.2.5).
g. Cable terminal dies, as specified (see 3.2.5.1).
h. Manual controls, if different (see 3.2.8). i. Coolant system, if different (see 3.2.9).
j. Coolant filtration system, if required (see 3.2.9).
k. Electrical system industrial standard, if different (see 3.2.11).
l. Electrical system voltage, as specified (see 3.2.11).
m. Drive motor enclosure, as specified (see 3.2.12).
n. Optional accessories, as specified (see 3.2.13).
o. Performance, if different (see 3.3).
p. Machine dimensions and weight, as specified (see 3.4).
q. System of units required (see 3.6).
r. Dual system of units, if required (see 3.6). s. Nameplate, if different (see 3.7).
t. Lubrication plate or chart, if required (see 3.8).
u. Hydraulic fluid plate or chart, if required (see 3.9).
v. Testing, if different (see 5.3).
w. Test responsibility, if different (see 5.3.1).
x. Certification of calibration, as specified (5.3.1).
y. Optional tests, as specified (see 5.3.5).
z. Acceptance, if different (see 5.4).
aa. Packaging requirements, if different (see 6).

7.3 Part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor.

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- Size: Sizes are shown in tables I, II, and III
- Type: A - Type I, Rotary die swaging machine
  B - Type II, Stationary die swaging machine
  C - Type III, Rotary die-closing swaging machine
- CID number

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MILITARY INTERESTS:
Custodians:
Army - AR
Navy - SH
Air Force - 99

Reviewer Activities:
Navy - MC, OS
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:
GSA - FAS

Preparing Activity:
DLA - GS6

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